Loops: for and while

Programming · Level 2 · Lesson 4



In this lesson you will

- Repeat with for and range.
- Count and add up in a loop.
- Use while, and avoid a loop that never ends.

Key ideas

- range(1, 6) gives 1 to 5.
- total = 0 before the loop.
- while needs something inside to change.

Activity: Be the loop

1. Warm-up (3 min): How would you print 1 to 100?
2. Act (7 min): One of you is i, one is total.
3. Predict (7 min): Change range; predict the total.
4. Run (3 min): Check your predictions on the phone.

Acting it out works without a phone; one phone is enough to check at the end.

Let's discuss

1. Why does range stop before its second number?
2. When would you use while instead of for?
3. How could a loop help you search a list?

Exit ticket

- Write a loop that prints 10, 20, 30, 40, 50.